

Module 2: Network + Network Security , Maintenance and Troubleshooting

Troubleshooting Tools

Task 1:

```
C:\Users\aasha>ipconfig /all

Windows IP Configuration

Host Name . . . . . : Aashanap
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
DNS Suffix Search List. . . . . : localdomain


Ethernet adapter Ethernet:

Connection-specific DNS Suffix . : localdomain
Description . . . . . : vmxnet3 Ethernet Adapter
Physical Address. . . . . : 00-0C-29-F5-35-A7
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::24a6:ba6:bdc8:973c%14(Preferred)
IPv4 Address. . . . . : 192.168.249.131(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : 10 August 2026 13:54:11
Lease Expires . . . . . : 10 August 2026 16:03:25
Default Gateway . . . . . : 192.168.249.2
DHCP Server . . . . . : 192.168.249.254
DHCPv6 IAID . . . . . : 100666409
DHCPv6 Client DUID. . . . . : 00-01-00-01-31-C3-AC-79-00-0C-29-F5-35-A7
DNS Servers . . . . . : 192.168.249.2
Primary WINS Server . . . . . : 192.168.249.2
NetBIOS over Tcpip. . . . . : Enabled
```

IP - 192.168.249.131

Default Gateway - 192.168.249.2

Task 2:

```
aashanapatel@192 ~ % ping www.google.com
PING www.google.com (142.251.150.119): 56 data bytes
64 bytes from 142.251.150.119: icmp_seq=0 ttl=118 time=28.407 ms
64 bytes from 142.251.150.119: icmp_seq=1 ttl=118 time=20.889 ms
64 bytes from 142.251.150.119: icmp_seq=2 ttl=118 time=15.444 ms
64 bytes from 142.251.150.119: icmp_seq=3 ttl=118 time=14.438 ms
64 bytes from 142.251.150.119: icmp_seq=4 ttl=118 time=15.167 ms
64 bytes from 142.251.150.119: icmp_seq=5 ttl=118 time=14.691 ms
64 bytes from 142.251.150.119: icmp_seq=6 ttl=118 time=14.955 ms
64 bytes from 142.251.150.119: icmp_seq=7 ttl=118 time=15.150 ms
64 bytes from 142.251.150.119: icmp_seq=8 ttl=118 time=14.921 ms
```

Task 3:

```
aashanapatel@192 ~ % traceroute www.flipkart.com
traceroute to flipkart.com (163.53.76.86), 64 hops max, 52 byte packets
 1  gpon.net (192.168.1.1)  4.576 ms  3.581 ms  3.243 ms
 2  * * *
 3  *
```

Hops = 64

Task 4:

```
aashanapatel@192 ~ % nslookup www.spotify.com
Server:          192.168.1.1
Address:         192.168.1.1#53

Non-authoritative answer:
www.spotify.com canonical name = atc.spotify.map.fastly.net.
Name:   atc.spotify.map.fastly.net
Address: 151.101.107.42
```

nslookup is a command-line tool used to query DNS (Domain Name System) servers and troubleshoot name resolution issues

Task 5:

ipconfig

Check this first to confirm your laptop actually has a valid IP address, subnet mask, and default gateway (not something like 169.254.x.x, an APIPA address, which would mean you're not even properly connected to the network). This rules out a basic local network/DHCP problem before looking further.

2. nslookup zomato.com

Once you know your laptop has a working IP configuration, check whether DNS resolution is working — is the domain name actually translating to an IP address? If this fails or times out, the problem is DNS-specific (Zomato's servers might be fine, but your machine can't find their address). This directly matches the earlier scenario you worked through: "can access by IP but not by name."

3. tracert zomato.com (or ping zomato.com if DNS resolved successfully)

If DNS resolved fine in step 2, use tracert to see the actual path packets take toward Zomato's server — this reveals *where* along the route the connection is failing or slowing down (e.g., it dies at your ISP's router, or somewhere further out on the internet). Alternatively, a simple ping here confirms whether the server is reachable at all and gives you round-trip time/packet loss, which is a quicker first check before running the more detailed tracert.